

Project Report: Nutritional Food Clustering Using K-Means Clustering

Abstract

This project presents an unsupervised machine learning approach for grouping food items according to their nutritional characteristics using K-Means Clustering. The analysis was conducted on the cleaned Nutrition Analytics dataset containing 1,609 food records and 11 original columns. Nine numerical nutritional features—Calories, Protein, Total Fat, Carbohydrates, Fiber, Sugar, Sodium, Cholesterol, and Potassium—were selected for clustering. Since these features are measured in different units and numerical ranges, StandardScaler was applied before clustering to ensure that each feature contributed on a comparable scale. To determine an appropriate number of clusters, candidate values of K from 2 to 10 were investigated using the Elbow Method and Silhouette Score. Solutions containing extremely small clusters were also avoided to maintain meaningful and interpretable groups. Based on these considerations, $K = 4$ was selected as the final number of clusters. The final model achieved an inertia of 7006.57 and a silhouette score of 0.4496, indicating a moderate or partly overlapping cluster structure. Analysis of the resulting nutrient profiles identified four data-driven groups: High-Carbohydrate / High-Sugar Foods, High-Fat / High-Calorie Foods, Lower-Calorie / Lower-Nutrient Foods, and High-Protein / High-Cholesterol Foods. Cluster centers, average nutrient values, cluster distributions, and standardized nutrient profiles were examined to understand the characteristics of each group. Principal Component Analysis (PCA) was additionally used to visualize the clustering result in two dimensions, while the K-Means model itself was trained using all nine standardized nutritional features. Overall, the project demonstrates how unsupervised machine learning can be used to discover meaningful nutritional patterns without requiring predefined target labels.

1. Introduction

Food products contain different amounts of nutrients such as calories, protein, total fat, carbohydrates, fiber, sugar, sodium, cholesterol, and potassium. Because these nutritional values vary considerably among foods, analyzing multiple nutrients together can provide a better understanding of their overall nutritional characteristics. However, manually examining a large number of food records and identifying similar nutritional profiles can be difficult and time-consuming. Machine learning techniques can help automate this process and reveal patterns within nutritional data. The main objective of this project is to identify groups of foods with similar nutritional characteristics using K-Means Clustering. The cleaned Nutrition Analytics dataset contains 1,609 food records, providing sufficient numerical information for discovering similarities among foods. Nine numerical nutrient variables are used as the input features for the clustering process. This project is an unsupervised learning problem because the dataset does not contain a predefined target variable representing nutritional groups. Instead of predicting an existing class, the algorithm discovers groups directly from similarities in the nutritional values. Food Name and Food Category are not used as input features for K-Means, because the clustering is intended to depend only on numerical nutritional characteristics. Food Category is used only after clustering to support the interpretation of the resulting groups. Before applying K-Means, the numerical features are standardized because they have different measurement units and ranges. The project then investigates different numbers of clusters using the Elbow Method and Silhouette Score. After selecting the final number of clusters, the resulting groups are analyzed using cluster sizes, cluster centers, average nutrient values, standardized nutrient profiles, and visualizations. PCA is also applied to provide a two-dimensional visual representation of the clustering result. Through this approach, the project aims to demonstrate how unsupervised machine learning can transform nutritional data into understandable groups and reveal differences among food items based on their nutrient composition.

2. Literature Review

Clustering is an unsupervised machine learning technique used to organize observations into groups based on similarity. Unlike supervised learning, clustering does not require predefined class labels or a target variable. The objective is to discover patterns or structures that already exist within the data. This makes clustering suitable for nutritional datasets where food items contain numerical nutrient information but may not have predefined nutritional profile labels. Among clustering techniques, K-Means Clustering is one of the most widely used partition-based algorithms for numerical data. The algorithm divides observations into a predefined number of clusters, represented by K . Initially, cluster centers or centroids are established, and each observation is assigned to the nearest centroid based on distance. The centroids are then updated using the observations assigned to each cluster. This assignment and updating process continues iteratively until the cluster assignments stabilize or the optimization process converges. The objective is to reduce the within-cluster sum of squares, also known as inertia. An important consideration when using K-Means is feature scaling. Nutritional variables can have very different numerical ranges and measurement units. For example, protein and carbohydrates are measured in grams, while sodium and cholesterol are measured in milligrams and energy is represented in kilocalories. Because K-Means is distance-based, variables with larger numerical scales could have a greater influence on the clustering process. Therefore, this project applies `StandardScaler` to transform the selected features to comparable standardized scales before training the model. Another important issue is selecting an appropriate value of K . The Elbow Method evaluates the inertia for different numbers of clusters. As K increases, inertia generally decreases because more clusters allow observations to be located closer to their respective centroids. The goal is to inspect the point where additional clusters provide progressively smaller reductions in inertia. The Silhouette Score provides another way to evaluate clustering quality. It considers both how closely an observation is related to its own cluster and how separated it is from other clusters. Higher silhouette values generally indicate better-defined clustering structure. In this project, both the Elbow Method and Silhouette Score are examined, together with cluster-size considerations, rather than selecting K from a single metric alone. Finally, PCA reduces the nine nutritional features into two principal components for visualization. It is used only for visualization, while K-Means is trained on all nine standardized features.

3. Related Work

Data analysis and machine learning techniques have increasingly been applied to food and nutrition data to identify nutritional patterns, compare food products, classify food items, and analyze similarities in nutrient composition. Nutrition databases contain multiple quantitative attributes for each food item, making them suitable for statistical analysis and machine learning applications. Structured nutrition resources such as Nutrition Value and USDA FoodData Central provide information about calories, macronutrients, minerals, and other nutritional properties of food products. Such structured data can be used to compare food items and investigate relationships among different nutritional variables. In this project, the cleaned Nutrition Analytics dataset provides a similar numerical structure suitable for unsupervised analysis. Previous machine learning approaches to food data may use supervised classification when known food categories or labels are available. In such cases, a model learns from labelled examples and attempts to predict a predefined category. However, this project addresses a different problem. It does not attempt to predict the existing Food Category column. Instead, the objective is to discover nutritional groups that emerge from the numerical nutrient values themselves. K-Means is therefore used to group foods according to similarities across nine nutritional features. The resulting cluster numbers initially have no nutritional meaning. Their characteristics are determined only after clustering by examining standardized nutrient profiles, average nutrient values, and cluster centers. This allows labels such as High-Carbohydrate / High-Sugar or High-Protein / High-Cholesterol to be assigned from the observed characteristics rather than being predefined before model training. This project therefore differs from traditional food classification by focusing on data-driven nutritional profiling through unsupervised clustering. The combination of K-Means, feature standardization, Elbow and Silhouette analysis, cluster-profile interpretation, and PCA visualization provides a complete workflow for discovering and explaining nutritional patterns within the dataset.

4. Methodology

The overall workflow was: Dataset Verification → Feature Selection → Standardization → K Selection → K-Means Clustering → Cluster Interpretation → PCA Visualization → Evaluation.

4.1 Dataset

The verified cleaned dataset contains 1,609 records and 11 columns, with no missing values and no exact duplicate rows. The clustered output also contains 1,609 records; two additional columns store the assigned cluster and its interpretation.

Property	Value
Total food records	1,609
Original columns	11
Numerical features used	9
Target variable	None
Learning type	Unsupervised
Algorithm	K-Means Clustering

4.2 Feature Selection and Standardization

The nine clustering features are Calories (kcal), Protein (g), Total Fat (g), Carbohydrates (g), Fiber (g), Sugar (g), Sodium (mg), Cholesterol (mg), and Potassium (mg). Food Name and Food Category are excluded from model training. StandardScaler is applied so that each nutrient contributes on a comparable standardized scale.

4.3 Selection of the Number of Clusters

K values from 2 to 10 were evaluated using the Elbow Method and Silhouette Score. Although $K = 7$ produced the highest raw silhouette score (0.4781), it created a very small cluster of only 2 foods, and several higher-K solutions also produced extremely small clusters. Therefore, $K = 4$ was selected as the best reasonably sized and interpretable solution, achieving a silhouette score of 0.4496.

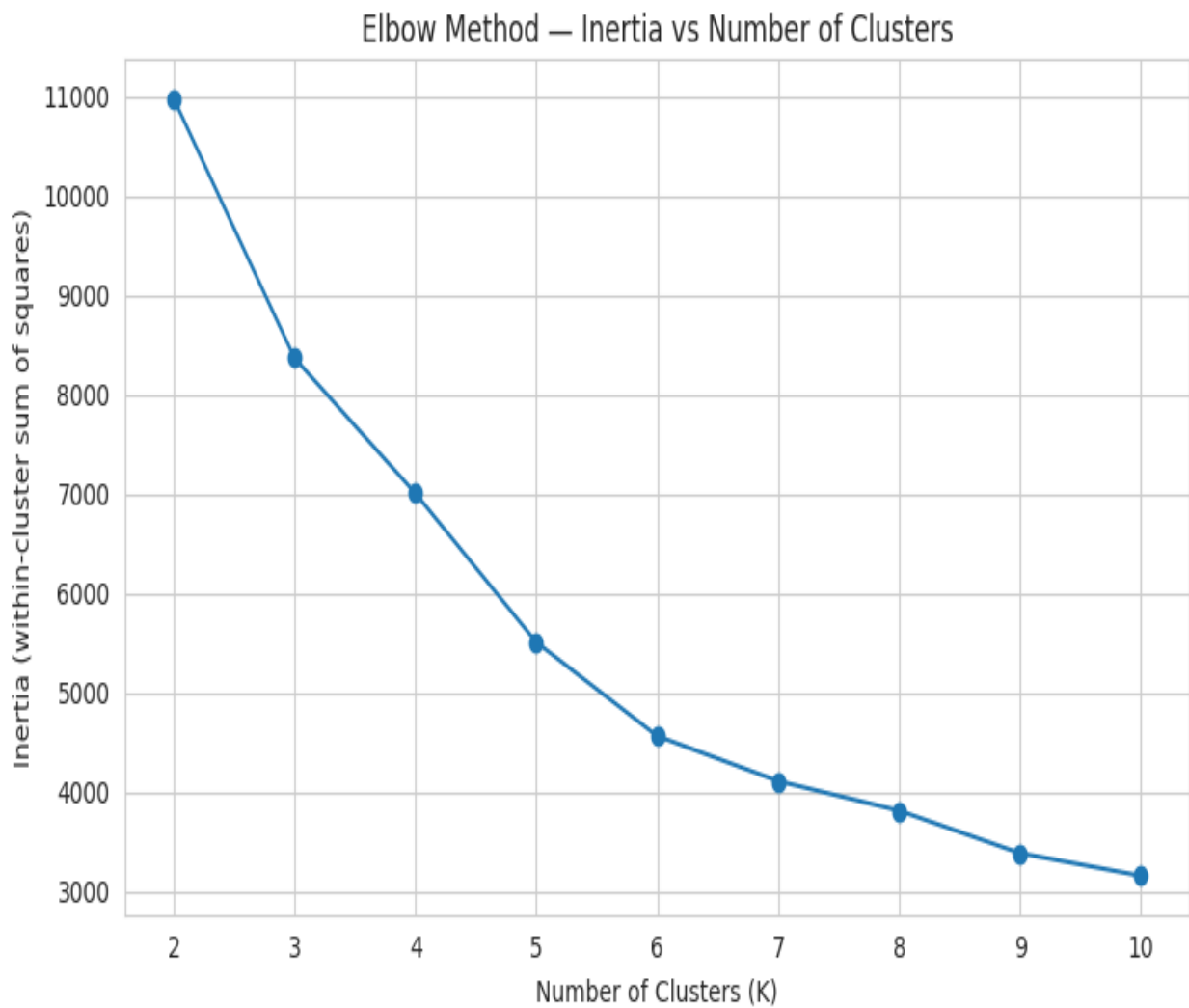


Figure 1: Elbow Method - Inertia vs Number of Clusters

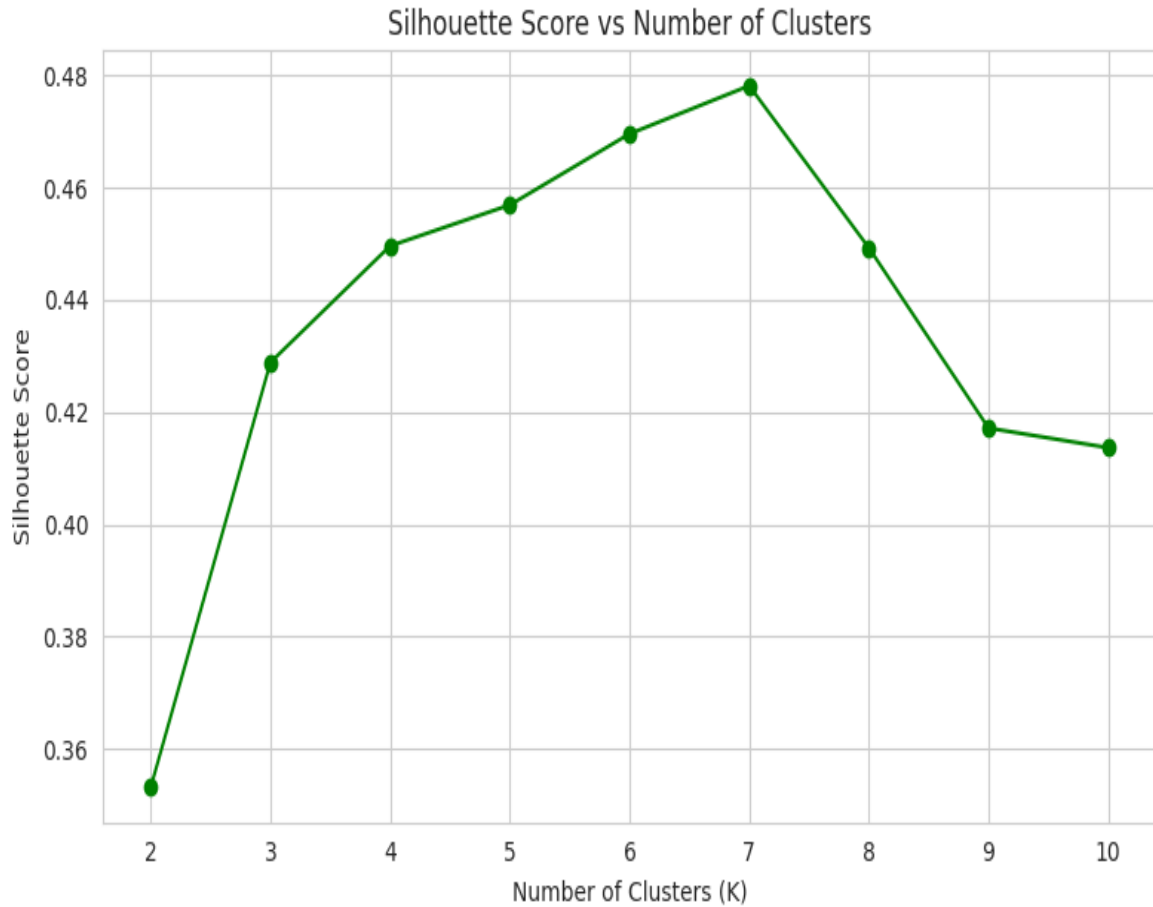


Figure 2: Silhouette Score vs Number of Clusters

4.4 Final K-Means Model and PCA

The final K-Means model uses $K = 4$, all nine standardized nutrient features, `random_state = 42`, and `n_init = 10`. PCA with two components is applied only after clustering to create a two-dimensional visualization. The two PCA components do not replace the original nine features used to train K-Means.

5. Result Analysis

5.1 Cluster Distribution

Cluster	Number of Foods	Percentage
0	310	19.27%
1	253	15.72%
2	774	48.10%
3	272	16.90%

Cluster 2 is the largest group with 774 foods (48.10%), while Cluster 1 is the smallest with 253 foods (15.72%). The cluster-size comparison is used descriptively to show how records are distributed across the four groups.

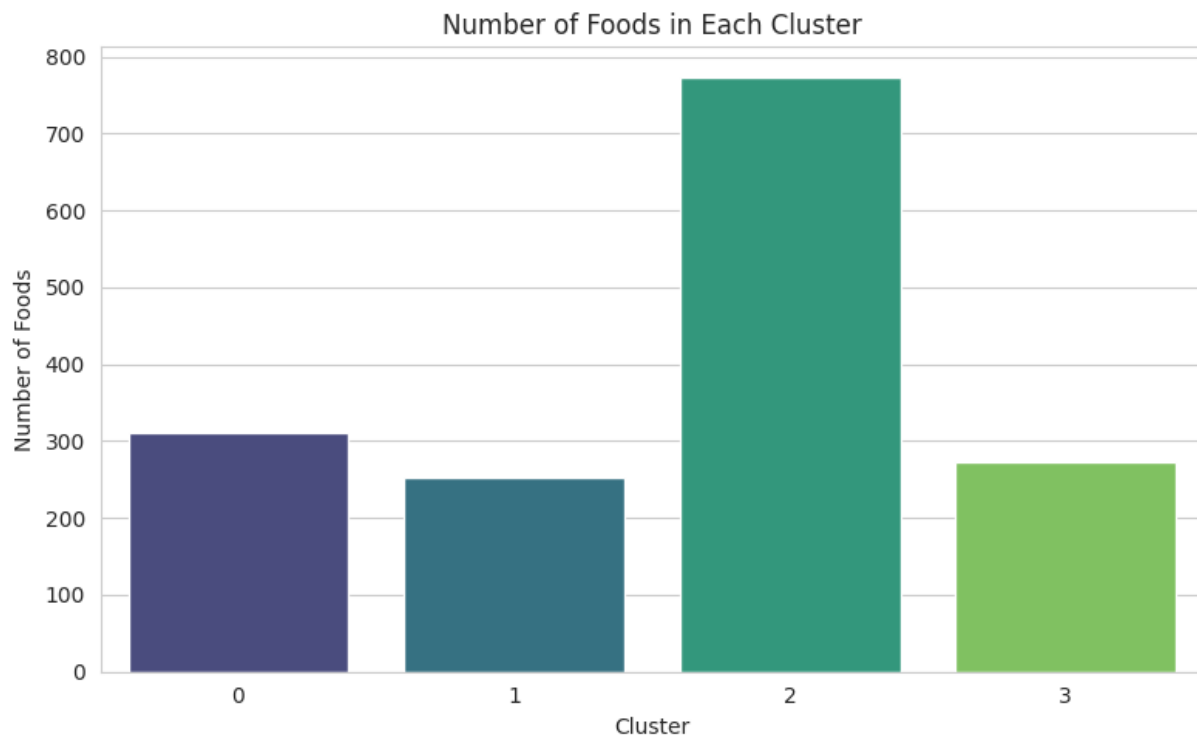


Figure 3: Number of Foods in Each Cluster

5.2 Average Nutritional Profile

Cluster	Calories	Protein	Fat	Carbs	Fiber	Sugar	Sodium	Chol.	Potassium
0	398.99	10.87	12.15	69.55	1.75	46.37	9.40	5.86	5.37
1	494.05	24.90	32.59	44.28	9.93	16.26	67.91	0.43	14.58
2	81.71	5.92	2.51	12.54	1.65	5.17	28.61	1.44	4.23
3	314.81	40.67	23.32	5.97	0.17	2.69	23.66	27.55	3.22

The average profiles show clear differences among the four groups. Cluster 0 has the highest average carbohydrate and sugar levels. Cluster 1 has the highest average calories and total fat. Cluster 2 has comparatively low values for several major nutrients. Cluster 3 has the highest average protein and cholesterol.

5.3 Cluster Interpretation

Cluster	Data-Driven Interpretation
0	High-Carbohydrate / High-Sugar Foods
1	High-Fat / High-Calorie Foods
2	Lower-Calorie / Lower-Nutrient Foods
3	High-Protein / High-Cholesterol Foods

The labels are not predefined classes. They are suggested after examining the standardized nutrient profile of each cluster. Positive standardized values indicate nutrients above the overall dataset average, while negative values indicate below-average levels.

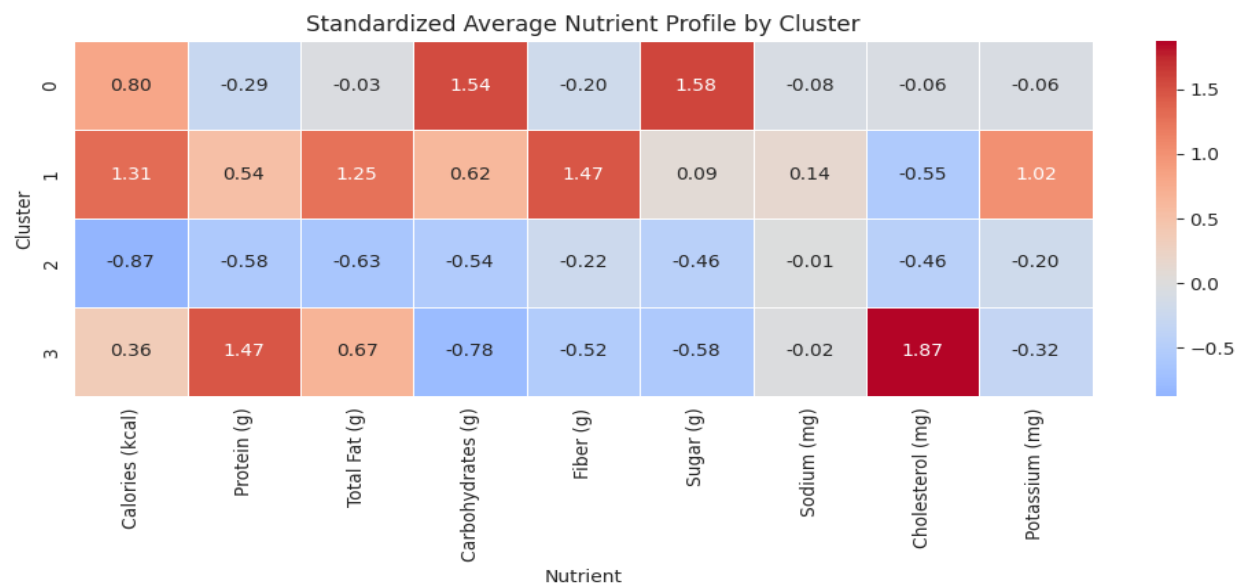


Figure 4: Standardized Average Nutrient Profile by Cluster

5.4 PCA Visualization

The PCA plot provides a simplified two-dimensional view of the nine-dimensional clustering result. Several groups show visible separation, while some observations overlap. This visual pattern is consistent with the moderate silhouette score and indicates meaningful but not perfect cluster separation.

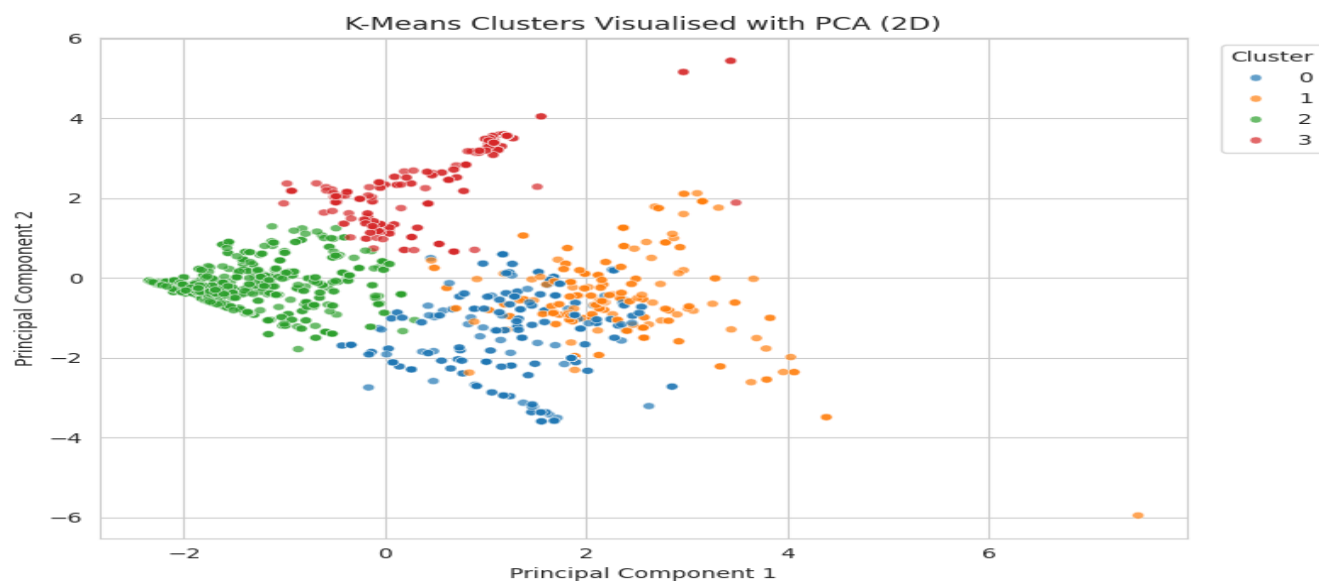


Figure 5: K-Means Clusters Visualized with PCA (2D)

5.5 Nutrient Comparison

The individual nutrient charts make the main cluster differences easier to interpret. Cluster 1 has the highest average calories (494.05 kcal/100 g) and total fat (32.59 g/100 g). Cluster 3 has the highest average protein (40.67 g/100 g), while Cluster 0 has the highest average carbohydrates (69.55 g/100 g).

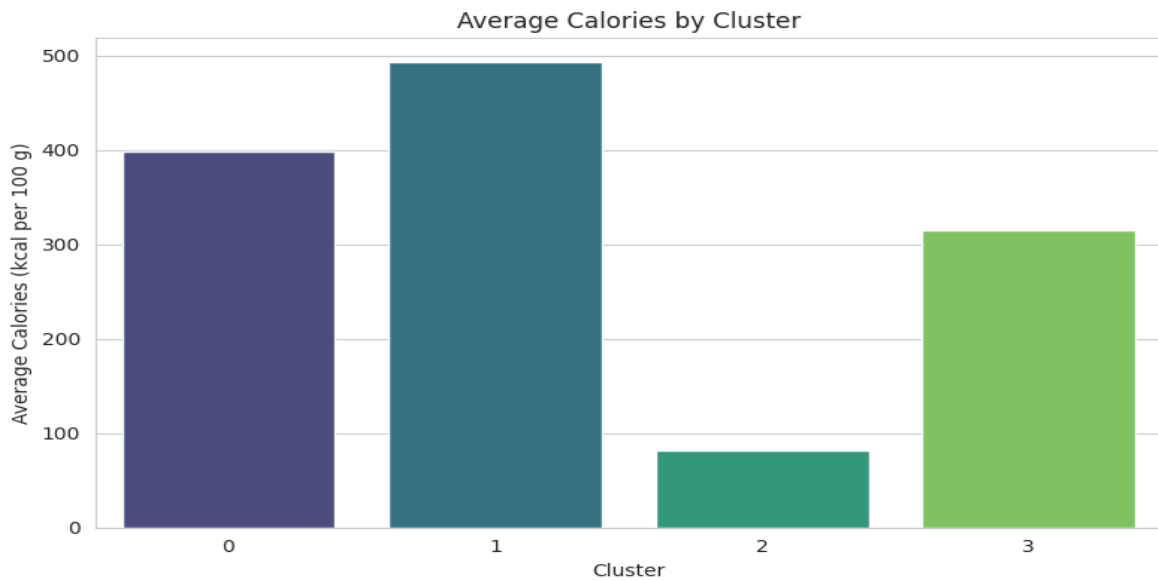


Figure 6: Average Calories by Cluster



Figure 7: Average Protein by Cluster

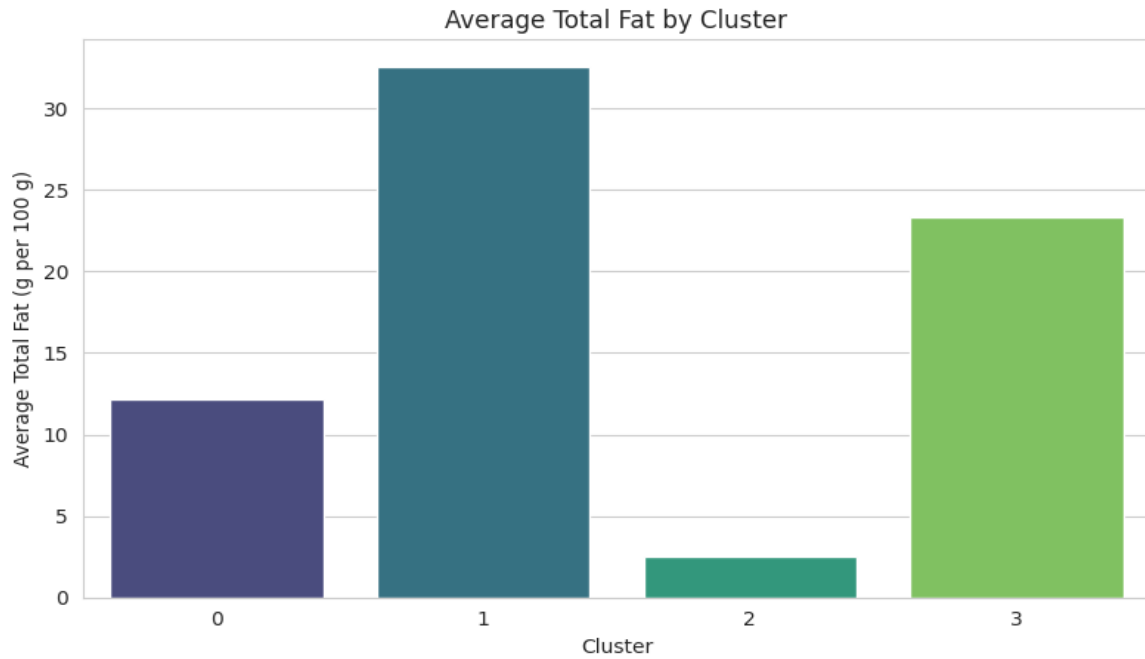


Figure 8: Average Total Fat by Cluster

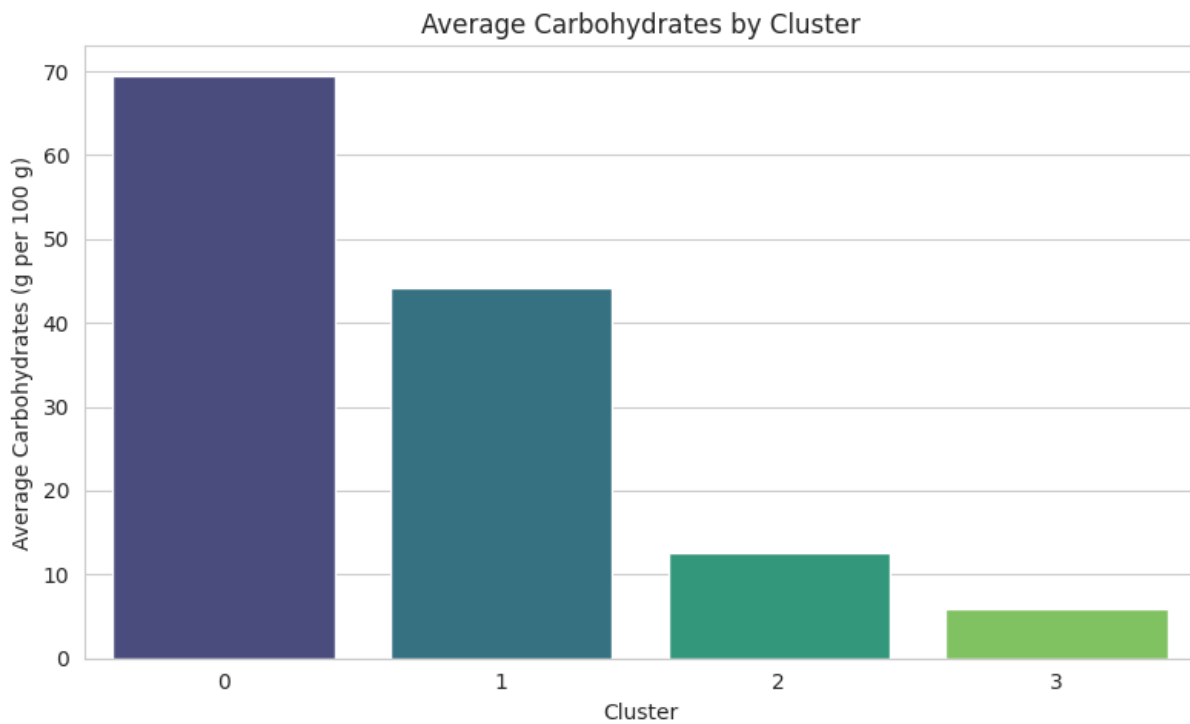


Figure 9: Average Carbohydrates by Cluster

5.6 Model Evaluation

K-Means is an unsupervised algorithm, so classification metrics such as accuracy, precision, recall, F1-score, and confusion matrix are not applicable. The final clustering is evaluated using inertia and the Silhouette Score.

Evaluation Metric	Result
Final K	4
Final Inertia	7006.57
Final Silhouette Score	0.4496
Interpretation	Moderate or partly overlapping cluster structure

The silhouette score of 0.4496 indicates moderate or partly overlapping cluster structure. Therefore, the model provides useful nutritional grouping, but the clusters should not be described as perfectly separated.

5.7 Key Findings

The analysis identifies four nutritional profiles. Cluster 0 is characterized mainly by high carbohydrate and sugar levels. Cluster 1 represents a high-fat and high-calorie profile. Cluster 2 contains nearly half of the records and shows comparatively lower calorie and nutrient levels. Cluster 3 has the strongest protein and cholesterol profile. Based on the variation among the standardized cluster profiles, Cholesterol (mg) was identified as the nutrient that most strongly distinguishes the clusters. The uploaded clustered CSV was cross-checked against a reproduced K-Means model using the same nine standardized features, $K = 4$, $\text{random_state} = 42$, and $n_init = 10$; all 1,609 cluster assignments matched.

6. Conclusion

This project successfully applied K-Means Clustering to 1,609 food records using nine standardized nutritional features. Candidate K values were investigated using the Elbow Method and Silhouette Score, together with a practical guard against extremely small clusters. The final selection of $K = 4$ achieved an inertia of 7006.57 and a silhouette score of 0.4496. The resulting clusters were interpreted as High-Carbohydrate / High-Sugar Foods, High-Fat / High-Calorie Foods, Lower-Calorie / Lower-Nutrient Foods, and High-Protein / High-Cholesterol Foods. PCA and nutrient comparison visualizations supported the interpretation. Overall, the project demonstrates how unsupervised learning can reveal meaningful nutritional patterns without requiring a predefined target variable.

References

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